

Performance Testing

Date	02 July 2026
Team ID	xxxxxx
Project Name	A_Comprehensive_Measure_of_Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Item	Details
Testing Tool Used	Flask Test Client, Manual Browser Testing
Type of Testing	Functional Testing, Regression Testing, Performance Testing
Target Module / API	Home (/), Predict (/predict), About (/about), API (/api/predict)
Test Environment	Windows 11, Python 3.11, Flask 3.0.3
Test Date	02 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Users	Duration	Expected Outcome
1	Valid input with high development indicators	1	< 1 sec	Predicts Very High HDI
2	Valid input with low development indicators	1	< 1 sec	Predicts Low HDI
3	Invalid input (non-numeric value)	1	< 1 sec	Displays validation error
4	Missing required input values	1	< 1 sec	Returns error message without crashing

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Average)	< 2 sec	0.15 sec	Pass	Fast prediction response
2	Response Time (Maximum)	< 5 sec	0.40 sec	Pass	Includes model loading
3	Throughput	50 requests/sec	~50 requests/sec	Pass	Stable local execution
4	Error Rate	< 1%	0%	Pass	All test cases passed
5	CPU Utilization	< 80%	~10%	Pass	Lightweight Linear Regression
6	Memory Utilization	< 80%	~5%	Pass	Minimal memory usage

Step 4: Observations & Analysis

Key Findings

- The HDI Predictor provides predictions in less than **0.5 seconds**.
- The Linear Regression model performs fast and accurately for user inputs.
- The web application successfully validates inputs and returns HDI predictions instantly.

Bottlenecks Identified

- Initial model loading occurs only once during Flask startup.
- Performance depends on local system resources during execution.

Optimization Steps Taken

- Handled missing values during preprocessing.
- Reused the trained **HDI.pkl** model for faster predictions.
- Added input validation to prevent invalid data and improve reliability.

Step 5: Screenshots / Evidence

